

Predicting 30 Day Hospital Readmission Using Prior Utilization EHR Diagnoses and Social Determinants of Health

A Leakage Safe Patient Level Evaluation

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Project scope

Prediction of all cause 30 day hospital readmission from index available prior utilization, principal diagnosis, and encounter linked SDOH data, followed by a separate diagnosis transition analysis among readmitted encounters.

Abstract

Thirty day hospital readmission is an important outcome for quality measurement and care planning, but accurate prediction is difficult because readmissions reflect clinical severity, prior use, social context, and the selected intervention threshold. This study developed a reproducible, leakage safe comparison of baseline utilization, electronic health record diagnosis, and social determinants of health feature blocks in a single hospital cohort. The analytic cohort contained 97,310 eligible index hospital encounters from 59,449 patients and 11,973 readmissions within 30 days. Patients were separated across training, validation, and held out test partitions with zero overlap. Four classifier families were evaluated across seven prespecified feature sets, yielding 28 directly comparable models. The strongest held out discrimination came from the full XGBoost model with AUROC 0.666 (patient bootstrap 95% CI 0.653 to 0.681), average precision 0.246, and Brier score 0.100. Diagnosis based EHR features contributed most of the improvement over the prior utilization baseline. SDOH alone had weak discrimination, but adding SDOH to baseline and EHR information produced a small incremental gain. At the validation selected Youden threshold, the full XGBoost model achieved 51.5% sensitivity, 74.2% specificity, and 21.6% precision. A separate post outcome analysis found that 42.8% of qualifying readmissions crossed to a different diagnosis body system. These findings support cautious use for risk stratification after external validation and workflow specific threshold selection, not autonomous clinical decision making. These results support the assertion that prediction models should be inclusive enough to mirror the multilayered complexity of a patient's external environmental barriers and investigate potential clinical comorbidities across body systems.

Keywords hospital readmission; social determinants of health; electronic health records; machine learning; patient level validation; clinical prediction

1 Introduction

Hospital readmissions within 30 days are used to assess outcomes after discharge and to identify opportunities for better transition planning. The Centers for Medicare and Medicaid Services established a hospital wide, all condition risk standardized readmission measure, making the 30 day interval a widely recognized quality window (Centers for Medicare & Medicaid Services, 2011). Emergency and inpatient reuse also matters operationally because repeated acute care often signals unresolved illness, limited continuity, or barriers outside the hospital. A frequent use threshold of four or more emergency visits has been used by health systems as a practical marker of concentrated service use (Health Quality Ontario, 2022).

Clinical records alone do not fully describe the circumstances that shape a return to care. Food insecurity, transportation barriers, housing instability, financial strain, social isolation, stress, and related social conditions can affect access to medications, follow up, and recovery. Standardized screening resources from CMS and LOINC make these constructs available in EHR data, although incomplete screening and nonresponse create substantial measurement challenges (Centers for Medicare & Medicaid Services, 2021; Regenstrief Institute, 2018; Chen et al., 2020).

This project asked whether SDOH information improves 30 day readmission prediction beyond prior acute care use and index diagnosis information. It also examined whether the diagnosis recorded at readmission repeated the index diagnosis or shifted to another clinical body system. The central design principle was comparability: every algorithm and feature block used the same eligibility rules, outcome, patient partitions, and held out test set.

2 Related Work

2.1 Readmission prediction and clinical baselines

Established readmission tools such as the LACE index summarize length of stay, acuity, comorbidity, and emergency use (van Walraven et al., 2010, as reviewed by Karam et al., 2023). Neural network studies have shown that large EHR feature sets can improve all cause readmission ranking, while also raising

questions about transportability and calibration (Jamei et al., 2017). More recent work has tested whether social variables improve conventional scores. Belouali et al. (2022) found that SDOH could improve the LACE index, and Ricket et al. (2024) evaluated social risk augmentation of the HOSPITAL score after acute myocardial infarction. These studies motivate nested comparisons in which the value of new data is measured against a defined clinical baseline.

2.2 SDOH in EHR prediction

A systematic review by Chen et al. (2020) found that EHR based SDOH variables are heterogeneous in definition, completeness, and predictive use. Zhang et al. (2020) reported that social determinants can affect models of potentially avoidable 30 day readmission or death, while Amrollahi et al. (2022) found improvement when SDOH was added to sepsis readmission models. Yang et al. (2023) similarly emphasized that predictive value depends on the care setting and the representation of social data. These findings support evaluating SDOH incrementally rather than assuming that social variables will be strong standalone predictors.

2.3 Dr Tchoua 2023 study and the present extension

Karam, MacCabe, and Tchoua (2023) studied emergency department readmissions using community health worker and SDOH information. Their work framed social information as both a predictive input and a potential link to community based intervention. The paper also situated prediction within broader concerns about inequity, representation, and bias in clinical machine learning (Karam et al., 2023). The present study follows that motivation but uses a distinct cohort and a stricter evaluation structure. It fixes one 97,310 encounter hospital cohort, separates patients rather than encounters, preserves unscreened encounters through explicit no record states, compares nested feature blocks, and evaluates the locked test set only after training, tuning, and threshold selection (Project Methods Log, 2026; Project Methods Log, 2026).

2.4 Measurement authority for SDOH domains

The feature engineering drew on established screening instruments and public authorities. The CMS Accountable Health Communities tool supports screening for health related social needs. The Hunger

Vital Sign provides a validated two item food insecurity screen (Hager et al., 2010). The PHQ 2 supports brief depression screening (Kroenke et al., 2003), while the Edinburgh Postnatal Depression Scale provides related evidence for perinatal contexts (Matijasevich et al., 2014). Stress interpretation was informed by Elo et al. (2003) and Wiegner et al. (2015). Physical activity measures were informed by the Exercise Vital Sign and federal activity guidelines (Coleman et al., 2012; Piercy et al., 2018). Financial strain, social participation, and contact frequency were interpreted using Tucher et al. (2024), Tomioka et al. (2018), and Fujiwara et al. (2022). These sources informed construct interpretation; they were not used to tune the outcome model.

2.5 Coding and data provenance

Principal diagnoses were standardized using the Agency for Healthcare Research and Quality Clinical Classifications Software Refined for ICD 10 CM (Agency for Healthcare Research and Quality, 2025). SDOH question provenance was checked against LOINC and CMS resources. Kansas Information for Communities documentation provided a public reference for Kansas hospital discharge data context, while the analytic patient level source was the local hospital EHR extract described in the project materials (Kansas Department of Health and Environment, n.d.;).

3 Methods

3.1 Study design and cohort

This retrospective prediction study used eligible index hospital encounters with complete 30 day follow up. The outcome was a subsequent hospital admission within 30 days after index discharge, consistent with the CMS measurement window (Centers for Medicare & Medicaid Services, 2011). The final cohort included 97,310 encounters from 59,449 patients, with 11,973 positive outcomes (12.30%). Readmission diagnosis was excluded from predictors to prevent future information from entering prediction.

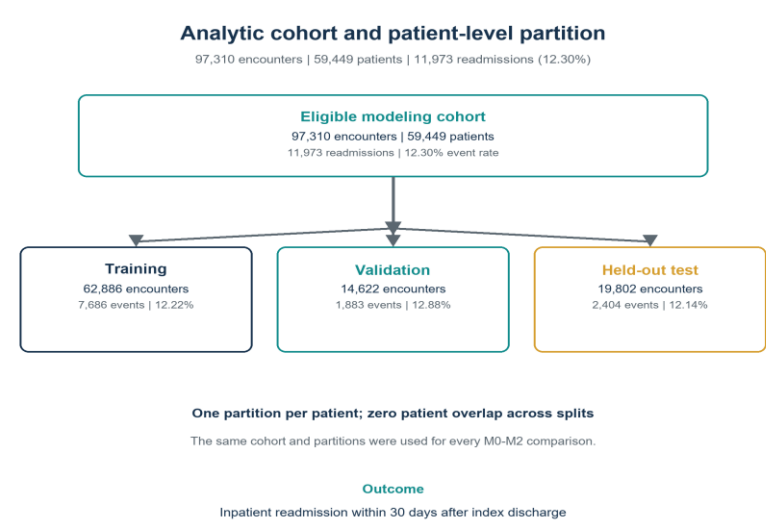


Figure 1. Analytic cohort construction
Legend. The same 97,310 encounters flowed into every comparison.

Partition	Encounters	Patients	Readmissions	Rate
Training	62,886	38,448	7,686	12.22%
Validation	14,622	8,926	1,883	12.88%
Held out test	19,802	12,075	2,404	12.14%
Total	97,310	59,449	11,973	12.30%

3.2 Predictor blocks

M0 contained one prespecified indicator for at least four unique ED or hospital encounters in the 365 days before the index encounter. The four encounter cutoff reflects the project definition of concentrated acute care use and is consistent with a published frequent use indicator (Health Quality Ontario, 2022; Project Methods Log, 2026). M1 added index principal diagnosis features mapped to AHRQ CCSR body systems with explicit mapping quality indicators (Agency for Healthcare Research and Quality, 2025). M2 added 12 SDOH domains derived from CMS and LOINC aligned screening constructs (Centers for Medicare & Medicaid Services, 2021; Regenstrief Institute, 2018).

Matrix	Definition	Tree features	Logistic features
M0	Prior high combined acute care use in previous 365 days	1	1
M1	M0 plus diagnosis body system and mapping quality	24	23
M2	M1 plus reduced mutually exclusive SDOH states	87	75

Each SDOH domain was reduced to mutually exclusive risk, clean negative, unable, other nonresponse or unresolved, no record, and when sufficiently common, conflict or partial states. This representation follows evidence that EHR social data completeness and recording processes can influence prediction (Chen et al., 2020; Yang et al., 2023). A conflict state was retained separately only when the development set contained at least 100 encounters and 20 readmission events. This project specific stability threshold was fixed before test evaluation and does not imply clinical equivalence between sparse and unresolved states (Project Methods Log, 2026). The test set did not influence category grouping or constant feature removal (Collins et al., 2015; Project Methods Log, 2026).

Stage	Inputs	Decision rule	Omitted from predictors	Final size
M0	Prior 365 day acute care use	High use = 4 or more distinct ED or hospital patient days; index day excluded	Raw counts, separate ED and hospital counts	1 predictor
M1	M0 plus index principal diagnosis	ICD 10 CM mapped to AHRQ CCSR body system; circulatory is logistic reference	Post index diagnoses and raw diagnosis text	24 tree; 23 logistic
M2	M1 plus 12 SDOH domains	One mutually exclusive state per domain; conflict retained at development $n \geq 100$ and events ≥ 20	Post index responses; clean negative indicators from logistic model	87 tree; 75 logistic

Table 1. Methodology matrix definitions and prespecified decisions from presentation (Project Methods Log, 2026).

3.3 Model development and evaluation

Logistic regression, histogram gradient boosting, random forest, and XGBoost were fitted to seven feature configurations. This nested design isolates the incremental information in diagnosis and SDOH while holding the denominator and partitions constant (Belouali et al., 2022; Amrollahi et al., 2022; Project Methods Log, 2026). Candidate hyperparameters were fitted on training patients and selected by validation AUROC. Median imputation occurred inside each pipeline. The selected estimator was refitted on training plus validation data and evaluated once on the held out test set (Collins et al., 2015; Project Methods Log, 2026).

Performance measures included AUROC, average precision, Brier score, calibration, and threshold dependent classification metrics. Average precision was included because the outcome prevalence was approximately 12%, making precision recall behavior informative in addition to ROC discrimination (Saito & Rehmsmeier, 2015). Uncertainty used 200 patient level bootstrap replicates to preserve within patient clustering (Project Methods Log, 2026). Thresholds

were fixed on validation data at 0.50, maximum Youden index, and sensitivity targets of at least 80% and 90%. Youden maximizes sensitivity plus specificity minus one (Youden, 1950); the sensitivity targets were prespecified workflow scenarios rather than universal clinical cutoffs (Project Methods Log, 2026; Project Methods Log, 2026).

3.4 Diagnosis transition analysis

Among the 11,973 readmitted index encounters, the next admission was classified hierarchically as exact ICD 10 CM match, same CCSR category, same body system, different body system, or unknown. The hierarchy was prespecified so each return occupied one mutually exclusive category (Agency for Healthcare Research and Quality, 2025; Project Methods Log, 2026). Time to return was compared with Kruskal Wallis followed by Dunn pairwise tests with Holm correction because the analysis compared multiple groups without assuming normal return times. The secondary model used index available M2 features and the original patient partitions (Project Methods Log, 2026; Project Methods Log, 2026).

3.5 Reproducibility safeguards

The final run used Python 3.12.13, pandas 3.0.5, scikit learn 1.9.0, XGBoost 3.4.0, and random seed 20260812. Patient overlap across partitions was zero. Identifiers, outcome fields, next admission information, days to readmission, complete follow up indicators, post index responses, and readmission diagnoses were excluded from predictors. The pipeline saved datasets, feature specifications, fitted models, metrics, plots, and QA logs.

4 Results

4.1 Held out model performance

The same ranking pattern appeared across all four classifier families. Baseline only AUROC was 0.569. EHR only models reached approximately 0.609 to 0.610, and baseline plus EHR reached 0.648 to 0.651. SDOH only models remained near 0.531 to 0.532. The full feature set achieved 0.661 to 0.666, with XGBoost highest.

The best held-out project AUROC of 0.666 falls within the range of the closest published studies reviewed for this project. Belouali et al. (2022) reported approximately 0.660, Zhang et al. (2020)

approximately 0.670, and Ricket et al. (2024) reported 0.635 to 0.672 across clinical, social-risk, and combined models, with a combined score near 0.669. This similarity supports the plausibility of the project estimate, but it is not a direct performance ranking because the studies used different populations, predictors, and outcome definitions.

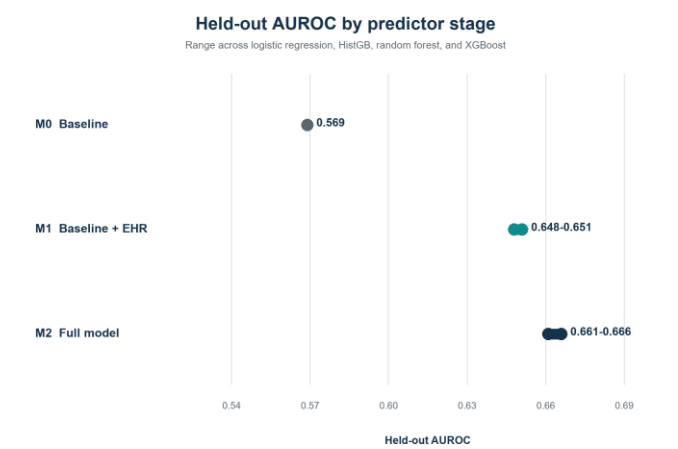


Figure 2. Held-out AUROC by predictor stage across four algorithms

Legend. M0 used prior high acute-care utilization, M1 added index EHR diagnosis, and M2 added SDOH. Horizontal ranges show the minimum and maximum held-out AUROC across logistic regression, histogram gradient boosting, random forest, and XGBoost.



Figure 3. Project AUROC in the context of closely comparable published results

Legend. The best project AUROC was 0.666. Published values provide context only because study populations, predictors, and readmission outcomes differed (Zhang et al., 2020; Belouali et al., 2022; Ricket et al., 2024).

Table 2. Held-out M0-M2 discrimination and validation-selected operating ranges

Stage	AUROC (95% CI envelope)	Recall	Precision
M0 Baseline	0.569 (0.556-0.584)	22.4%	26.6%
M1 Baseline + EHR	0.648-0.651 (0.633-0.666)	40.6-43.0%	22.8-23.5%
M2 Full model	0.661-0.666 (0.644-0.681)	44.8-51.5%	21.0-23.7%

Note. AUROC confidence intervals are envelopes spanning the lowest lower bound and highest upper bound among the four algorithm-specific patient-bootstrap intervals. Recall and precision are ranges at validation-selected Youden thresholds. The project presentation labels these stages M0-M2; there is no separate M3 stage.

Detailed Incremental ROC Curves

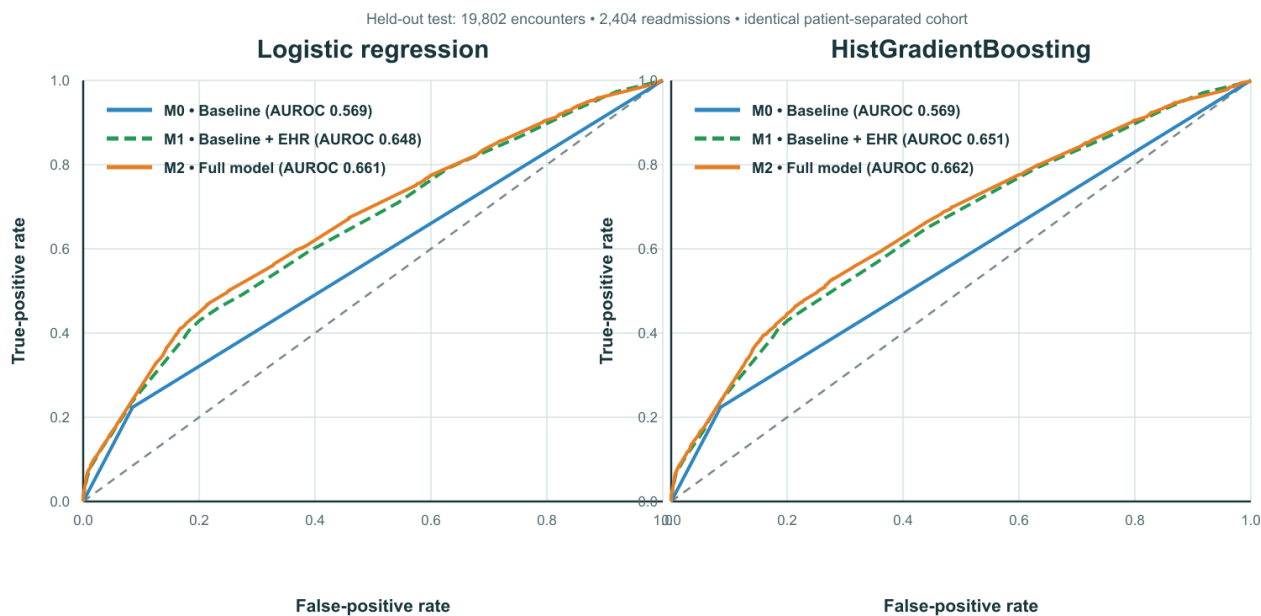


Figure 4. Incremental ROC curves for logistic regression and histogram gradient boosting

Legend. Held-out test $n=19,802$ with 2,404 readmissions. Each panel compares M0 baseline, M1 baseline plus EHR, and M2 full model. AUROC confidence intervals use patient-bootstrap estimates; recall and precision use validation-selected Youden thresholds.

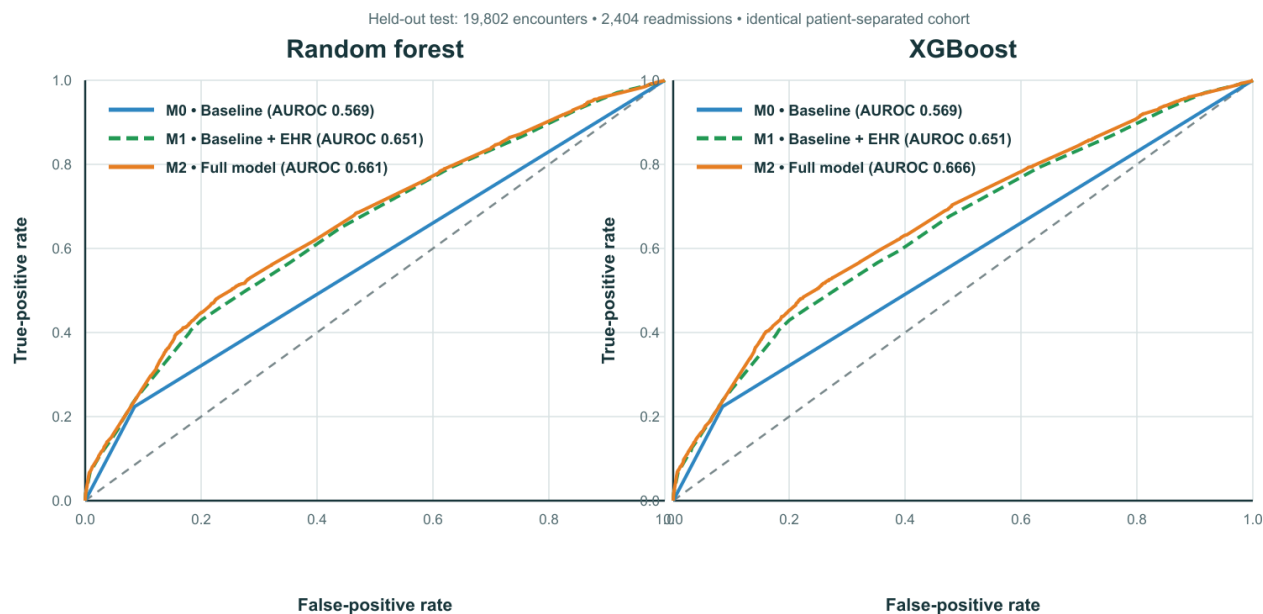


Figure 5. Incremental ROC curves for random forest and XGBoost

Legend. Held-out test n=19,802 with 2,404 readmissions. The XGBoost M2 model achieved the highest held-out AUROC, 0.666 (95% CI 0.653-0.681), with recall 51.5% and precision 21.6% at the validation-selected Youden threshold.

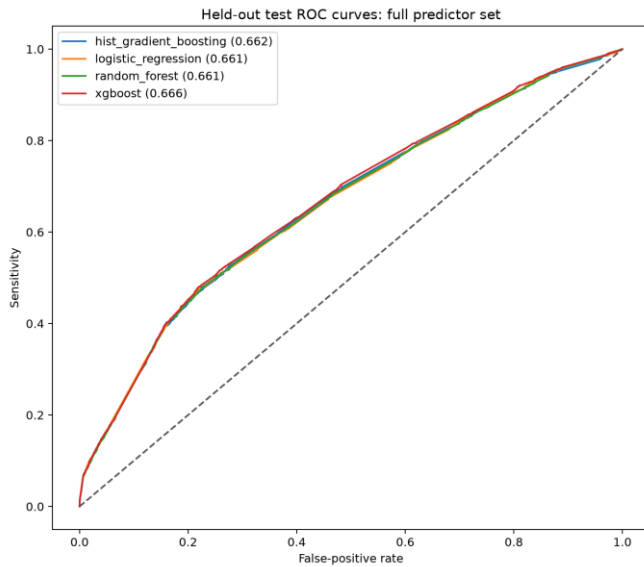


Figure 6. ROC curves for the full models

Legend. Sensitivity is plotted against one minus specificity. The diagonal represents chance.

Source: presentation results and heldout_test_model_comparison.csv.

Feature set	Logistic	HistGB	Random forest	XGBoost
Baseline only	0.569	0.569	0.569	0.569
EHR only	0.610	0.609	0.609	0.609
SDOH only	0.532	0.531	0.531	0.531
Baseline plus SDOH	0.586	0.586	0.585	0.587
Baseline plus EHR	0.648	0.651	0.651	0.651
EHR plus SDOH	0.631	0.633	0.631	0.635
Full	0.661	0.662	0.661	0.666

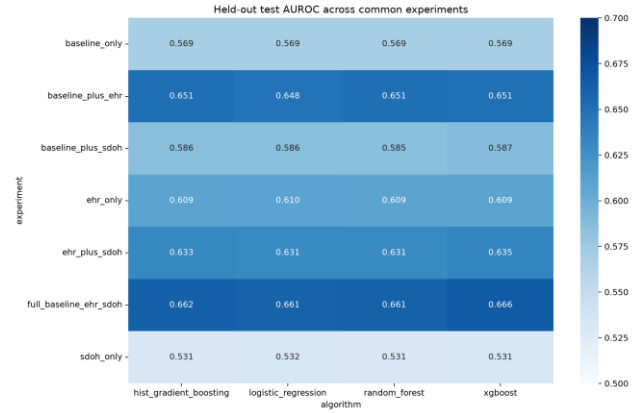


Figure 7. Held out AUROC across algorithms and feature blocks

Legend. Rows are feature blocks and columns are algorithms; darker cells indicate higher AUROC.

Source: heldout_test_model_comparison.csv.

4.2 Feature importance and clinical code context

Mean held-out permutation importance across logistic regression, histogram gradient boosting, random forest, and XGBoost showed that prior utilization and diagnosis context supplied the largest performance contributions. Permuting the prior high combined-utilization indicator reduced AUROC by 0.0500 on average, and permuting the pregnancy-related diagnosis indicator reduced AUROC by 0.0412. Food-insecurity no-record status (0.0165) and perinatal diagnosis (0.0134) were the next largest contributors. These values are mean changes in model AUROC after feature disruption; they are not individual-feature AUROCs, statistical significance tests, or causal effects.

Table 3. Consensus top 20 features by mean held-out permutation importance

Rank	Feature	Feature family or clinical-code context	Mean Δ AUROC
1	Prior high combined utilizer (365 d)	Utilization	0.0500
2	Pregnancy-related diagnosis	CCSR body system: Pregnancy	0.0412
3	Food insecurity - no record	SDOH documentation	0.0165
4	Perinatal diagnosis	CCSR body system: Perinatal	0.0134
5	Physical activity risk	SDOH measured risk	0.0071
6	Diagnosis missing	Diagnosis quality	0.0064
7	Depression - no record	SDOH documentation	0.0060
8	IPV - no record	SDOH documentation	0.0047
9	Transportation - no record	SDOH documentation	0.0044
10	Utilities - no record	SDOH documentation	0.0042
11	Body system unclassified	CCSR body system: Unclassified	0.0035
12	Smoking - nonresponse	SDOH documentation	0.0032
13	Mental/behavioral diagnosis	CCSR body system: Mental/behavioral	0.0029
14	Financial strain - no record	SDOH documentation	0.0028
15	Smoking - no record	SDOH documentation	0.0024
16	Factors influencing health status	CCSR body system: Factors/contacts	0.0021
17	Nervous system diagnosis	CCSR body system: Nervous system	0.0018
18	Neoplasm diagnosis	CCSR body system: Neoplasms	0.0017
19	Alcohol - no record	SDOH documentation	0.0017
20	Stress - no record	SDOH documentation	0.0010

Clinical-code note. The `ccsr_system_*` predictors are broad AHRQ CCSR body-system indicators derived from the index principal ICD-10-CM diagnosis. They do not represent detailed CCSR diagnosis categories or isolated ICD-10-CM code effects. The prominence of pregnancy and perinatal systems should therefore be interpreted as body-system context in this cohort, not as evidence that a specific code causes readmission. Several SDOH contributors are no-record or nonresponse states, which may reflect documentation processes as well as patient social risk.

4.3 Full model performance and threshold behavior

The full XGBoost model produced AUROC 0.666 (95% CI 0.653 to 0.681), average precision 0.246 (95% CI 0.225 to 0.274), and Brier score 0.1005 (95% CI 0.0968 to 0.1045). Calibration intercept was -0.002 and slope was 1.008. At the validation Youden cutoff of 0.1084, sensitivity was 51.5%, specificity 74.2%, precision 21.6%, and accuracy 71.5%. Lower thresholds increased sensitivity but generated many additional false positives.

The held-out readmission prevalence was 12.14% (12.30% overall). The full XGBoost model had average precision 24.6%, while precision at the validation-selected Youden threshold was 21.6%. Thus, the threshold-selected group had about 1.78 times the readmission concentration expected from an untargeted group of the same size (21.6% / 12.14%), and average precision was about 2.03 times the prevalence reference (24.6% / 12.14%). This is practically meaningful enrichment over the base rate, but it is not by itself a statistical-significance test and does not mean that 24.6% of all readmissions were identified; recall measures the proportion identified.

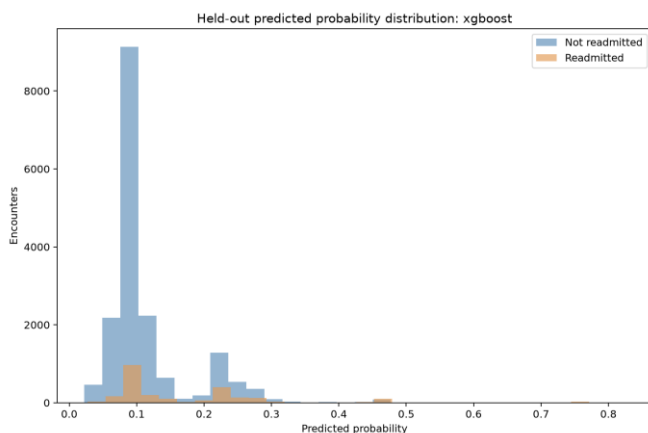


Figure 8. Full XGBoost predicted probability distribution

Legend. Event and nonevent scores overlap.

Threshold markers show sensitivity and false alert tradeoffs. Source:

heldout_test_metrics_all_thresholds.csv.

Rule	Cutoff	Sensitivity	Specificity	Interpretation
Fixed probability	0.500	2.0%	100.0%	Misses nearly all events
Validation Youden	0.108	51.5%	74.2%	Balanced statistical operating point

Rule	Cutoff	Sensitivity	Specificity	Interpretation
80% sensitivity target	0.092	79.4%	38.4%	More alerts and false positives
90% sensitivity target	0.084	86.0%	27.0%	Highest alert burden

4.4 Diagnosis relationship at readmission

Relationship	n	Percent	Median days	IQR days
Exact ICD 10 CM match	2,045	17.1%	13	6 to 21
Same CCSR category	1,315	11.0%	12	5 to 20
Same body system	3,078	25.7%	9	4 to 17
Different body system	5,128	42.8%	14	7 to 21
Unknown	407	3.4%	1	0 to 5

Time to readmission differed across the four known relationship groups (Kruskal Wallis $H(3) = 346.54$, $p < 0.001$). Five of six Dunn Holm adjusted comparisons were significant; exact ICD and same CCSR return times did not differ after correction (adjusted $p = 0.052$). Among known relationships, all four algorithms predicted a different body system return with similar held out AUROC, 0.808 to 0.810. This secondary target describes the type of return among patients already readmitted and must not be interpreted as readmission risk.

Table 4. Return timing and leakage-safe diagnosis-relationship prediction results

Analysis	Group or model	n	Estimate (95% CI)	Recall	Precision
Time to readmission	Different body system	5,128	Median 14 days	-	-
Time to readmission	Exact ICD-10-CM	2,045	Median 13 days	-	-
Time to readmission	Same CCSR category	1,315	Median 12 days	-	-
Time to readmission	Same body system	3,078	Median 9 days	-	-
Prediction	Random forest	2,320	AUROC 0.810 (0.789-0.829)	97.4%	62.2%
Prediction	HistGradientBoosting	2,320	AUROC 0.810 (0.788-0.830)	95.3%	63.2%
Prediction	Logistic regression	2,320	AUROC 0.809 (0.786-0.829)	86.1%	66.3%
Prediction	XGBoost	2,320	AUROC 0.808 (0.786-0.827)	95.9%	62.6%

Note. Time-to-readmission groups differed overall (Kruskal-Wallis $H(3)=346.54$, $p<0.001$). Five of six Holm-adjusted comparisons were significant; exact ICD-10-CM versus same CCSR was not significant (adjusted $p=0.052$). The leakage-safe prediction analysis used a held-out sample of 2,320 known diagnosis relationships with 45.4% positive prevalence. Recall and precision use validation-selected Youden thresholds. No formal pairwise AUROC test was performed.

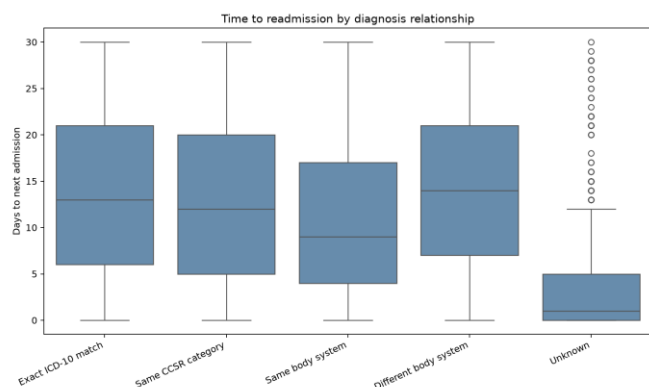


Figure 9. Time to readmission by diagnosis relationship

Legend. Same body system returns had the shortest known category median. Source: diagnosis relationship outputs.

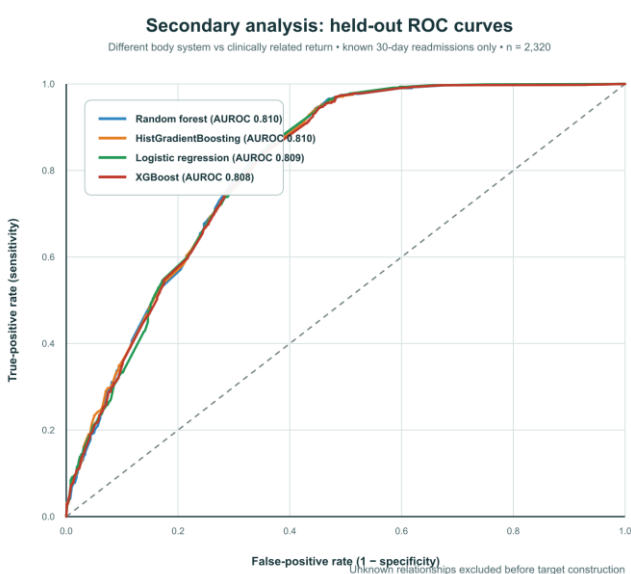


Figure 10. Secondary ROC curves for diagnosis system change

Legend. AUROC ranged from 0.808 to 0.810 among known readmission relationships; this is not overall readmission risk. Source: diagnosis_relationship_model_results.csv.

5 Discussion

The primary finding is that index diagnosis information supplied most of the incremental discrimination over a simple prior utilization baseline. SDOH alone performed only slightly above chance, yet SDOH contributed a small gain when combined with baseline and EHR information. The result agrees with prior work showing that social information can improve readmission models while rarely replacing clinical predictors (Zhang et al., 2020; Belouali et al., 2022; Amrollahi et al., 2022). It also supports the emphasis

of Karam et al. (2023) on using social information to guide human intervention rather than treating a risk score as an end in itself.

The modest AUROC does not imply that the model lacks all operational value, but it limits the appropriate use. The validation Youden threshold identified roughly half of subsequent readmissions and generated more false positives than true positives. A hospital that can offer a low risk intervention may accept a high alert volume. A resource intensive intervention would require a different operating point, stronger predictors, or a narrower target population. Threshold choice therefore belongs to the clinical workflow and cannot be selected from AUROC alone.

Explicit treatment of SDOH missingness is a methodological contribution. Unscreened, unable, declined, negative, and conflicting responses do not carry the same meaning. Preserving these states allowed the model to retain the full cohort instead of restricting analysis to selectively screened encounters. This approach reduces a major source of selection bias, although it cannot eliminate documentation bias or establish causal effects of social conditions.

The diagnosis transition analysis adds a complementary clinical observation: most readmissions did not exactly repeat the index principal diagnosis, and 42.8% crossed body systems. This heterogeneity helps explain why a narrow disease recurrence model may miss substantial all cause reuse. It also suggests that discharge planning and post discharge surveillance may need to account for broader patient vulnerability, not only recurrence of the index condition.

6 Limitations

- The study used a single health system and a deterministic patient separated split rather than temporal or external validation. Performance may change across sites, time periods, and documentation systems.
- The outcome was all cause hospital readmission. Planned admissions and service line specific exclusions were not part of the primary definition.
- SDOH measurements were limited to documented EHR screening. Missingness, selective screening,

response interpretation, and changes over time may introduce bias.

- Hyperparameter grids were intentionally bounded for reproducibility rather than exhaustive optimization.
- Bootstrap intervals preserved patient clustering, but subgroup fairness, decision curve net benefit, and prospective workflow effects remain unevaluated.
- The results are predictive and associational. They do not show that an SDOH exposure caused readmission or that changing the recorded exposure would prevent it.

7 Conclusion

A leakage safe patient level comparison showed that prior utilization, index diagnosis, and encounter linked SDOH information together produced the strongest 30 day hospital readmission model, although discrimination remained moderate. Diagnosis features drove most of the gain, while SDOH added a small increment and supplied important context about screening and access. The full XGBoost model should be considered a candidate risk stratification tool only after external or temporal validation, subgroup review, prospective calibration, and selection of a threshold matched to intervention capacity. The separate diagnosis analysis showed that readmissions frequently crossed clinical body systems, suggesting the rationale to consider clinical comorbidities in prediction models and reinforcing the need for broad transition planning rather than diagnosis recurrence alone.

8 Reproducibility and Project Materials

The technical narrative, methodology tables, and figures were reconciled only against slides 1 through 32 of the final project presentation and the underlying project outputs cited on those slides. The attached reference report controlled the compact academic layout, typography, caption style, and restrained use of color; its research content was not imported (Project Methods Log, 2026).

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